A Robust Feature Downsampling Module for Remote-Sensing Visual Tasks

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***Abstract*— Remote-sensing (RS) images present unique challenges for computer vision (CV) due to lower resolution, smaller objects, and fewer features. Mainstream backbone networks show promising results for traditional visual tasks. However, they use convolution to reduce feature map dimensionality, which can result in information loss for small objects in RS images and decreased performance. To address this problem, we propose a new and universal downsampling module named robust feature downsampling (RFD). RFD fuses multiple feature maps extracted by different downsampling techniques, creating a more robust feature map with a complementary set of features. Leveraging this, we overcome the limitations of conventional convolutional downsampling, resulting in a more accurate and robust analysis of RS images. We develop two versions of the RFD module, shallow RFD (SRFD) and deep RFD (DRFD), tailored to adapt to different stages of feature capture and improve feature robustness. We replace the downsampling layers (DSL) of existing mainstream backbones with the RFD module and conduct comparative experiments on several public RS image datasets. The results show significant improvements compared to baseline approaches in RS image classification, object detection, and semantic segmentation. Specifically, our RFD module achieved an average performance gain of 1.5% on the NWPU-RESISC45 classification dataset without utilizing any additional pretraining data, resulting in state-of-the-art performance on this dataset. Moreover, in detection and segmentation tasks on dataset for object detection in aerial images (DOTA) and instance segmentation in aerial images dataset (iSAID), our RFD module outperforms the baseline approaches by 2%–7% when utilizing pretraining data from NWPU-RESISC45. These results highlight the value of the RFD module in enhancing the performance of RS visual tasks. The code is available at https://github.com/lwCVer/RFD.**

***Index Terms*— Classification, detection, feature downsample, remote sensing (RS), segmentation.**

I. Introduction

CONVOLUTIONAL neural networks (CNNs) have been the primary approach for feature extraction from images in the past few years, following the remarkable success of AlexNet [1] in image classification [2]. Recently, the vision transformer (ViT) [3] has popularized transformer-based methods in computer vision (CV). The pyramid vision transformer (PVT) [4] leverages progressively shrinking pyramids to enhance model performance, while dramatically reducing the computational load required by the transformer to process large input feature maps. This approach has gained widespread adoption due to its efficient hierarchical backbone, which divides the network into downsampling layers (DSLs) and feature learning layers (FLLs).

Downsampling is the process of reducing the spatial resolution of feature maps in a backbone by performing operations such as maxpooling or strided convolutions. It is often used in backbones to increase the receptive field of neurons in deeper layers and to reduce the computational cost of the network. In most hierarchical backbone networks, DSLs commonly utilize convolution with a stride of 2 to reduce the size of feature maps. Although this method can effectively integrate local information, it has limitations in preserving fine-grained details, often resulting in the blurring or loss of features. This issue is especially prominent in remote-sensing (RS) images, where smaller and more complex objects require detailed information for accurate visual analysis. Merely downsampling feature maps via convolution with a stride of 2 may result in the loss of critical object features, which are crucial for subsequent feature learning, thereby leading to a negative impact on network performance.

To address this issue, we propose a new and general downsampling module called robust feature downsampling (RFD) to compensate for the deficiencies of convolutional downsampling in RS images. Our RFD module extracts more robust downsampled feature maps by fusing features from multiple downsampling methods, enabling subsequent FLLs to better capture key information and improve the overall performance of the network. Fig. 1 shows excerpts of downsampled feature maps at different scales of Swin V2 tiny [5] init and Swin V2 tiny + RFD. The first and third rows represent the feature maps of Swin V2 tiny init, while the second and fourth rows represent the feature maps of Swin V2 tiny + RFD. The multiples of downsampling are represented by 4×, 8×, 16×, and 32×, respectively. The models were trained using the training set of NWPU-RESISC45 [6], and the test images were randomly selected from the validation set of NWPU-RESISC45. For display purposes, we selected features that are more visually recognizable. As illustrated in the red box in Fig. 1, in the 4× and 8× feature maps perceptible

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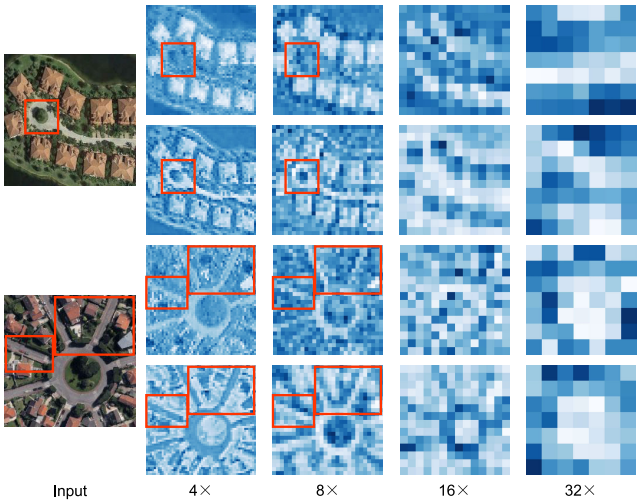


Fig. 1. Visualization of feature map of Swin V2 tiny init and +RFD models. The input images include dense residential and roundabouts. The first and third rows are excerpts of downsampled feature maps at different scales of Swin V2 tiny init. The second and fourth rows are excerpts of downsampled features at different scales for Swin V2 tiny + RFD. 4×, 8×, 16×, and 32× represent multiples of downsampling, respectively. The test images are randomly selected from the validation set of NWPU-RESISC45. For display purposes, we selected features that are friendly to human vision.

to human vision, the +RFD model retains more texture and detail information than the initial model.

In short, the RFD method involves replicating the initial feature representation into three copies, denoted as X, C, and M, during a DSL. Subsequently, these replicated features are then subjected to downsampling using three different techniques, and the resulting feature maps are fused to produce a consistent and robust representation. X performs convolutional downsampling to fuse local information of the feature map and improve generalization. C performs cut-slice downsampling to preserve the original data information of the feature map. M performs maxpooling downsampling to avoid the loss of key features and retain detailed information on the feature map. The resulting downsampled feature maps are concatenated and reduced to one-third dimensions of channels for increased robustness.

By merging multiple feature maps, our RFD method overcomes the limitations of convolutional DSLs and excels in capturing intricate details, resulting in a more robust feature representation. This approach not only enhances model performance, but also introduces a successful improvement solution for analyzing image characteristics in RS imagery analysis, thereby advancing the field.

We propose two variants of the RFD module, the shallow RFD (SRFD) module and the deep RFD (DRFD) module for shallow and deep DSLs, respectively. They are used to replace the DSLs in various types of hierarchical backbone networks (e.g., CNN, transformer, multi-layer perceptron (MLP), and the mixture of CNN and transformer). This verifies the versatility of the RFD module. We evaluate the proposed RFD module on RS image classification, object detection, and semantic segmentation tasks. The experimental results demonstrate that models with the RFD module significantly outperform the initial models. The RFD module can be used to replace the DSLs in different hierarchical backbone networks,

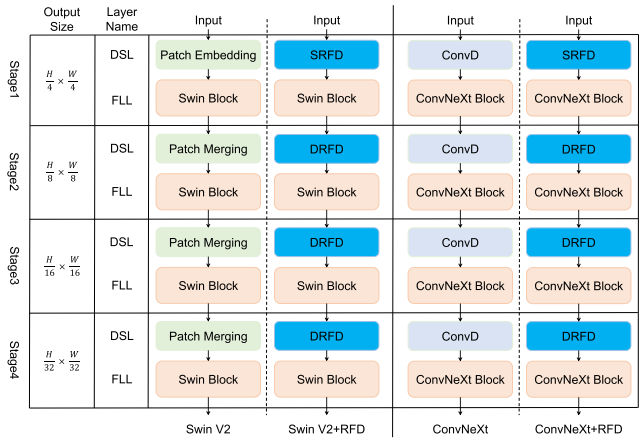


Fig. 2. Illustration of the proposed RFD module replacing DSLs in swin transformer V2 and ConvNeXt models. "+RFD" represents that the DSLs of the model are replaced by the RFD module. The SRFD module and the DRFD module are used to replace the stage 1 DSL and stage 2–4 DSLs of the original backbone networks, respectively. ConvD denotes convolutional downsampling. DSL denotes the downsampling layer. FLL denotes the feature learning layer.

such as swin transformer V2 [5] and ConvNeXt [7], as shown in Fig. 2. The SRFD module replaces the stage 1 DSL, while the DRFD module replaces the stage 2–4 DSLs of the original backbone networks.

Overall, this article contributes to the development of more effective and efficient methods for processing RS imagery by introducing a novel downsampling module specifically designed for RS visual tasks. The proposed RFD method is versatile and can be applied to a range of different backbone network architectures, making it suitable for a range of applications such as classification, detection, and segmentation. Specifically, our results indicate an average improvement of 1.5% across various RFD models for the classification task, an average improvement of 3.5% for the detection task, and an average improvement of 5% for the segmentation task. The contributions of this work are summarized as follows.

1) The proposed RFD method addresses the challenge of processing RS imagery with low resolution and small objects and effectively improves the accuracy of deep-learning models for RS image analysis.
2) The SRFD and DRFD modules based on RFD are designed to replace DSLs in different types of hierarchical backbone networks, demonstrating the versatility of the RFD module.
3) The RFD method is evaluated on RS image classification, object detection, and semantic segmentation tasks, and experimental results show that it significantly improves the performance of the models compared to initial models.

This is a summary of the sections in the article. Section I introduces the RFD method for feature downsampling, while Section II outlines the RS vision tasks, backbone networks of various types, and associated downsampling methods. Section III designs and explains the SRFD module and the DRFD module that are for shallow and deep DSLs, which are used to replace the DSLs in various types of hierarchical backbone networks. Section IV presents the experimental

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results of RFD for three RS vision tasks, followed by an ablation study of the method. Section V concludes the article and provides an outlook on future work.

II. Related Work

A. RS Visual Tasks

RS image analysis plays a vital role in geosciences, and visual tasks are of great interest to researchers in this field. Scene classification is one such task, and researchers have used a variety of datasets to explore this task. The UC Merced land use (UCM) [8] dataset is a popular choice for scene classification, as it contains a diverse set of images with varying levels of complexity. The NWPU-RESISC45 [6] dataset is another widely used dataset for this task, containing 45 different land-use classes. The AID [9] dataset contains 30 different land-use classes, and the EuroSAT [10] dataset contains 10 different land-use classes. Finally, the Million-AID [11] dataset is one of the largest and most diverse datasets for scene classification, containing over one million images across 50 different land-use classes. These datasets have been used by researchers to develop and evaluate new methods, such as [12], [13], [14], for scene classification.

Another important visual task is object detection in RS image analysis, which involves identifying and localizing objects of interest, such as buildings, vehicles, and people, in RS images using CV and machine-learning techniques. The widely used datasets for object detection include HRSC2016 [15] dataset that contains high-resolution optical images of objects such as ships, planes, and cars. The dataset for object detection in aerial images (DOTA) [16] dataset that contains aerial images of objects such as cars, ships, and buildings. The SpaceNet [17] dataset that contains high-resolution satellite imagery of objects such as buildings and roads. Researchers have explored various methods on these datasets, such as [18], [19], [20].

Furthermore, semantic segmentation is indeed a crucial task in RS image analysis. It enables the identification and mapping of different land-cover types, which can be used for a variety of applications, such as monitoring urbanization, managing natural resources, and assessing environmental changes. The ISPRS Vaihingen and Potsdam datasets are two widely used benchmark datasets for semantic segmentation in RS image analysis, and they contain high-resolution aerial images with detailed annotations of land-cover types. The instance segmentation in aerial images dataset (iSAID) [21] is another recently released dataset that contains more diverse and complex scenes, which pose additional challenges for segmentation algorithms. Researchers have proposed various methods [22], [23], [24], [25], for semantic segmentation, such as deep-learning-based approaches, graph-based methods, and hybrid techniques that combine multiple algorithms.

Visual tasks are an essential component of RS, attracting considerable attention from researchers in recent years. With the rise of technology, many advancements have been made to improve the accuracy and efficiency of these tasks. Researchers have focused on developing new methods and tools to enhance image classification, object detection, and segmentation techniques. These advancements have enabled RS applications to be more effective in a variety of fields, such as environmental monitoring, agriculture, and urban planning. Further research is expected to continue driving the development of new visual tasks and improved accuracy and efficiency in the field as the demand for RS applications grows.

B. Backbone Networks

1) Convolution Backbones: The proposal of AlexNet [1] revolutionized the field of CV, ushering in a new era of CNNs. Since then, numerous CNN-based networks have been developed, such as very deep convolutional networks for large-scale image recognition (VGG) [26], GoogleNet [27], ResNet [28], and RegNet [29].

In most backbone networks, DSLs are implemented using a convolutional layer with stride 2. For example, the stem-cell layer (shallow feature DSL) in the standard ResNet [28] includes a 7×7 convolution with stride 2, followed by a 3×3 maxpooling with stride 2, resulting in a $4\times$ downsampling of the input images. The spatial DSL (deep feature DSL) is achieved by the residual block at the start of each stage and uses 3×3 convolution with stride 2 (and 1×1 convolution with stride 2 at the shortcut connection). ConvNeXt [7] uses a 4×4 convolution with stride 4 in shallow feature DSL and uses a 2×2 convolution with stride 2 in deep feature DSLs.

2) Transformer Backbones: Since the introduction of ViT [3], the development of ViT architectures has experienced a significant boost in research. One notable advancement is PVT [4], which utilizes a progressive shrinking pyramid to reduce computations of large feature maps. This hierarchical backbone divides the network into DSLs and FLLs and has been widely adopted in other backbone networks such as the swin transformer [30] and CSwin [31], among others.

Conventionally, feature DSLs use convolution as a general method. For instance, CSwin uses a 7×7, stride 4 convolution for shallow feature DSLs and a 3×3 convolution with stride 2 for deep feature DSLs. In contrast, SSA [32] utilizes a 7×7, stride 2 convolution, a 2×2, stride 2 convolution for shallow feature DSLs, adds a $n \times (3 \times 3)$, stride 1 convolution between the two convolutional layers (n varies with model size), and uses a 3×3 convolution with stride 2 for deep feature DSLs.

3) Mixture of CNNs and Transformer Backbones: There are a mixture of CNNs and transformer backbones, such as CoAtNet [33] and Mixformer [34]. They also use convolution for DSLs. As an example, the Mixformer network uses twice 2×2, stride 2 convolution for shallow feature DSL and adds $2 \times (3 \times 3)$, stride 1 convolution between the two convolutional layers. Additionally, the network uses a 2×2 convolution with stride 2 for deep feature DSLs.

4) MLP Backbones: There are many MLP-based backbone networks such as MLP-mixer [35], an axial shifted MLP architecture for vision (AS-MLP) [36], and Hire-MLP [37]. AS-MLP [36] uses a 4×4, stride 4 convolution for shallow feature DSL and uses a cut-slice downsampling for deep feature DSLs.

5) Global Filter Backbones: GFNet-H [38] learns long-term spatial dependencies in the frequency domain with log-linear complexity. It uses a 4×4, stride 4 convolution for shallow feature downsampling and uses a 2×2 convolution with stride 2 in deep feature DSLs.

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C. Downsampling Modules

Downsampling is a fundamental operation in the CV that reduces the resolution of an image or feature map while preserving its important content. Downsampling helps to improve the efficiency of neural network models, reduce overfitting, and improve generalization performance. In this section, we will provide an overview of some of the downsampling modules in the CV.

Maxpooling [1] is a widely used downsampling module that selects the maximum value within a predefined window in the feature map. Maxpooling is computationally efficient and helps to reduce the dimensionality of the feature maps. However, maxpooling can result in the loss of some important spatial information and can lead to the overfitting of the model. Average pooling is similar to maxpooling but selects the average value within a predefined window in the feature map. Average pooling is also computationally efficient and helps to reduce the dimensionality of the feature maps. However, like maxpooling, average pooling can result in the loss of some important spatial information. Strided convolution [27] is a commonly used downsampling module that uses convolutional filters with a stride larger than 1 to reduce the resolution of the feature map. Strided convolution can preserve more spatial information than pooling and has been shown to be effective in some applications. LIP [39] proposed local importance-based pooling whose pooling layer is based on local importance modeling. Antialiased CNNs [40] showed that when integrated correctly, low-pass filtering can be compatible with existing architectural components such as maxpooling and strided convolution and can increase classification accuracy. SoftPool [41] maintained the functionality of the pooling layer while minimizing the loss of information during pooling.

In summary, downsampling modules are essential in the CV and play a critical role in reducing the dimensionality of feature maps while preserving important spatial information. The choice of the downsampling module depends on the specific application, and researchers are constantly exploring new and improved methods to improve the performance of CV models.

III. Methodology

In this section, an overview of the proposed RFD method is first presented to present an overview of the whole framework of this model in Section III-A. To strengthen the different stages of the feature capture ability of the RFD, we then introduce the SRFD and DRFD modules. We provide details of each submodule in Sections III-B and III-C. Finally, the combination of SRFD and DRFD on backbone networks is interpreted in Section III-D.

A. Overview of the +RFD Backbone

RS imagery poses unique challenges for CV tasks due to the lower resolution, smaller objects, and fewer features present in these images. To address this issue, we propose a novel and versatile downsampling method called RFD, which combines multiple feature maps extracted through various downsampling techniques to produce a more robust feature map with a complementary set of features, to address the unique challenges of RS imagery for CV tasks, such as lower resolution, smaller objects, and fewer features present in these images.

In brief, the RFD method involves replicating the initial feature representation into three distinct copies—X, C, and M—during a DSL. These replicated features then undergo downsampling via three distinct techniques and are fused to produce a consistent and robust feature representation. X performs convolutional downsampling to fuse local information of the feature map and improve generalization by preventing the model from memorizing the exact pixel positions of features. C performs cut-slice downsampling to preserve the original data information of the feature map. M performs maxpooling downsampling to avoid the loss of key features and retain detail information by selecting maximum values within each rectangle. The resulting downsampled feature maps are concatenated and reduced to one-third dimensions of channels for more robust feature maps.

We developed two versions of the RFD module, namely SRFD and DRFD, tailored to adapt to different stages of feature extraction and improve the robustness of the extracted features. The backbone structure after replacing the DSLs with the proposed RFD module is depicted in Fig. 3(a). The SRFD module replaces the DSL of stage 1 (shallow downsampling), which the feature map size from $H \times W$ to $(H/4) \times (W/4)$. The DRFD module replaces the DSLs of stages 2–4 (deep downsampling), in which the input size of DSL from $H \times W$ to $(H/2) \times (W/2)$ every time. Fig. 3(b) and (c) provides details for the SRFD and DRFD modules, respectively.

B. Shallow RFD

When an initial image is input into a backbone network, it typically contains a vast amount of information and redundancy at the pixel level. Therefore, to enhance the performance, it is essential to selectively preserve salient features while suppressing redundant information from the raw pixel images. Although most backbone networks use a 4×4 or 7×7, stride 4 convolutional layer to downsample the original size to $4\times$, this downsampling method is not very effective for the characterization of RS images. It leads to the loss of features from small RS objects and adversely affects the final performance. To overcome this issue, we propose an SRFD module based on the RFD approach, which aims to efficiently represent shallow feature maps. The process of SRFD is shown in Fig. 3(b).

In numerous previous studies, such as ConvNext [7], it has been demonstrated that the 7×7 convolution kernel is effective in fusing local features and filtering redundant information, thus facilitating the extraction and filtration of initial RS image information. Here, we apply a 7×7 convolution with stride 1 to the input image, resulting in a feature map that strengthens critical features while filtering redundant information. This process is referred to as the feature reinforcement layer, which increases the channel count from 3 to 1/4C and plays an important role in the subsequent processing steps.

Following the feature reinforcement layer, the resulting feature map x is duplicated into y_1 and y_2. We downsample y_1

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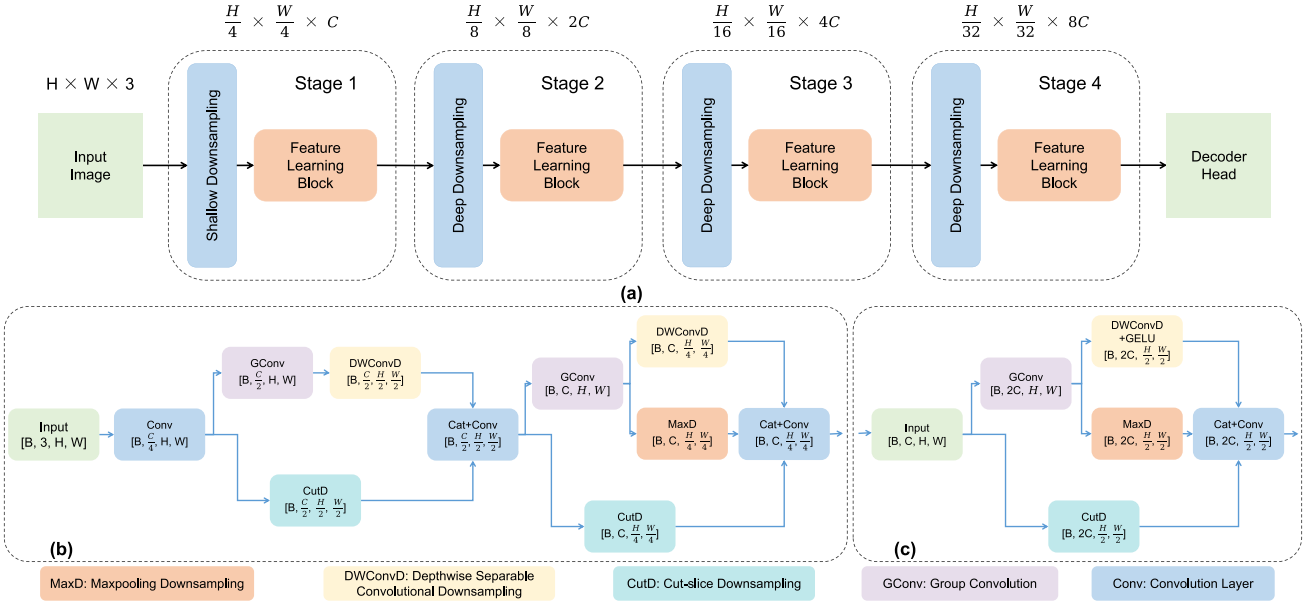


Fig. 3. Overall architecture of the proposed RFD. (a) General structure after replacement with SRFD and DRFD modules. (b) and (c) Details of SRFD and DRFD modules, respectively.

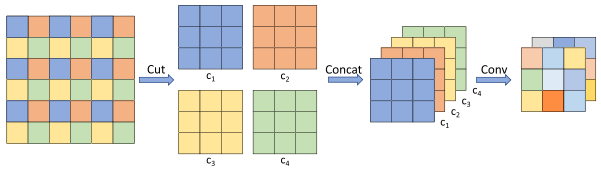


Fig. 4. Cut-slice downsampling. Cut denotes slicing adjacent pixel points into c_1, c_2, c_3, and c_4. Concat denotes concatenating feature maps $\{c_1, c_2, c_3,$ and $c_4\}$. Conv denotes a 1×1 convolution with stride 1 and the number of channels reduces from 4C to 2C.

using the cut-slice method, as illustrated in Fig. 4. This technique effectively reduces the feature map size while preserving the original feature information by slicing adjacent pixel points, yielding a matrix with half the original dimensions. Specifically, the resulting matrix consists of the following elements:

$$c_1 = \begin{bmatrix} x_{11} & x_{13} & \cdots \\ x_{31} & x_{33} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}_{\frac{H}{2}\times\frac{W}{2}}, \quad c_2 = \begin{bmatrix} x_{12} & x_{14} & \cdots \\ x_{32} & x_{34} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}_{\frac{H}{2}\times\frac{W}{2}}$$

$$c_3 = \begin{bmatrix} x_{21} & x_{23} & \cdots \\ x_{41} & x_{43} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}_{\frac{H}{2}\times\frac{W}{2}}, \quad c_4 = \begin{bmatrix} x_{22} & x_{24} & \cdots \\ x_{42} & x_{44} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}_{\frac{H}{2}\times\frac{W}{2}}$$

where x_{ij} is the feature located at the position of (i, j). The number of channels increases from 1/4C to C by concatenating feature maps c_1, c_2, c_3, and c_4, which reduces from C to 1/2C by 1×1 convolution with stride 1. The downsampling process thus enhances computational efficiency while maintaining critical features for subsequent analysis. As shown in (2), we name it $\boldsymbol{D_{cut}}$. To facilitate representation, we define a function called $\boldsymbol{fusion}$ that concatenates multiple features and fuses them into the desired number of channels. This function is used to efficiently combine and integrate diverse feature representations, as shown in (1).

The processing of y_2 starts with a 3×3 group convolution with a stride of 1, where the group number equals the number of input channels and the output channels are twice the number of input channels. This operation is followed by downsampling using a 3×3 depthwise convolution with a stride of 2. The depthwise convolution integrates local feature information, thereby improving feature fusion and reducing the feature size by half. By leveraging group convolution and depthwise convolution, the proposed approach reduces the network's floating-point operations (FLOPs) compared to traditional convolutional downsampling. We have named this approach $\boldsymbol{D_{conv}}$ and represent it mathematically in (3)

$$\boldsymbol{fusion} = \mathrm{BN}(\mathrm{Conv}(\mathrm{Concat}(x, y, z))) \tag{1}$$

$$\boldsymbol{D_{cut}} = \boldsymbol{fusion}(c_1, c_2, c_3, c_4) \tag{2}$$

$$\boldsymbol{D_{conv}} = \mathrm{BN}(\mathrm{DWConvD}(\mathrm{GConv}(x))) \tag{3}$$

where BN, Conv, DWConvD, and GConv denote batch normalization, convolution, depthwise convolution, and group convolution, respectively. Then, the number of channels in the output feature map is increased to C by concatenating y_1 and y_2. After this step, the number of channels is reduced from C to 1/2C using a 1×1 convolutional layer with stride 1, as described in the following equation:

$$y_1 = \boldsymbol{D_{cut}}(x), \quad y_2 = \boldsymbol{D_{conv}}(x)$$
$$\boldsymbol{X} = \boldsymbol{fusion}(y_1, y_2) \tag{4}$$

where Concat denotes concatenation.

During the downsampling process from the original size to a $2\times$ reduction, careful consideration must be given to the shallowest feature maps, as they tend to contain a significant amount of noise. The addition of maxpooling downsampling at this stage may result in the retention of such noise, which can have a negative impact on subsequent FLLs. Experimental results show that, in the original size to $2\times$ DSL, the SRFD

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module's accuracy is higher without the inclusion of maxpooling downsampling. Hence, we opt to exclude maxpooling downsampling from this stage.

In the process of downsampling the input feature map $\boldsymbol{X}$ by a factor of 2× to 4×, three methods are performed on $\boldsymbol{X}$ to obtain the output feature maps x_1, x_2, and x_3. First, x_1 is obtained by $\boldsymbol{D_{conv}}$, resulting in a reduction in spatial dimension by a factor of 2× while increasing the number of channels from 1/2 C to C. Second, x_2' is generated by performing group convolution on $\boldsymbol{X}$ using parameters shared with those of $\boldsymbol{D_{conv}}$. Similar to x_1, the number of channels in x_2' is increased from 1/2 C to C. Afterward, a 2 × 2 maxpooling operation with a stride of 2 is applied to x_2', resulting in feature downsampling. That x_2 is obtained. We name it $\boldsymbol{D_{max}}$ as shown in (5). x_3 is downsampled by $\boldsymbol{D_{cut}}$, and channels increase from 1/2 C to C

$$\boldsymbol{D_{max}} = \mathrm{BN}(\mathrm{maxpooling}(\mathrm{GConv}(\boldsymbol{X}))). \tag{5}$$

Then the number of channels becomes 3C by concatenating x_1, x_2, and x_3. The number of channels of the output feature maps $\boldsymbol{Y}$ reduces from 3C to C by 1 × 1, stride 1 convolutional layer. As shown in the following equation:

$$\begin{aligned} x_1 &= \boldsymbol{D_{conv}}(\boldsymbol{X}) \\ x_2 &= \boldsymbol{D_{max}}(\boldsymbol{X}) \\ x_3 &= \boldsymbol{D_{cut}}(\boldsymbol{X}) \\ \boldsymbol{Y} &= \boldsymbol{fusion}(x_1, x_2, x_3). \end{aligned} \tag{6}$$

C. Deep RFD

During the feature extraction process, the output feature map generated from the FLLs of stage 1 undergoes further processing in stages 2, 3, and 4. Before entering the FLL of stages 2, 3, and 4, the size of the feature map will be reduced to half of the input size. Since the semantic information of the feature map at these stages is rich, it becomes essential to avoid losing semantic information during downsampling. In conventional approaches, downsampling is accomplished using convolutional layers, but this method is unsuitable for the semantic information-rich feature maps, as it can alter or even discard essential features.

To overcome this limitation, we propose a novel solution called the DRFD module. The architecture of the DRFD module is illustrated in Fig. 3(c). Similar to the SRFD's 2× to 4× DSL, the input feature map $\boldsymbol{Y}$ is downsampling by $\boldsymbol{D_{conv}}$, $\boldsymbol{D_{cut}}$, and $\boldsymbol{D_{max}}$ to obtain the output feature maps x_1, x_2, and x_3, respectively. x_1 is obtained by $\boldsymbol{D_{conv}}$ meanwhile with a Gaussian error linear units (GELUs) [42] activation function, and the number of channels increases from C to 2C. x_2 is downsampled using $\boldsymbol{D_{cut}}$, and the number of channels increases from C to 2C. x_3 is downsampled using $\boldsymbol{D_{max}}$, and the number of channels increases from C to 2C, as demonstrated in the following equation:

$$\begin{aligned} x_1 &= \mathrm{GELU}(\boldsymbol{D_{conv}}(\boldsymbol{Y})) \\ x_2 &= \boldsymbol{D_{cut}}(\boldsymbol{Y}) \\ x_3 &= \boldsymbol{D_{max}}(\boldsymbol{Y}). \end{aligned} \tag{7}$$

Then the number of channels of feature map becomes 6C by concatenating x_1, x_2, and x_3. The number of channels of output feature map $\boldsymbol{X}$ reduces from 6C to 2C by 1 × 1, stride 1 convolutional layer, as shown in the following equation:

$$\boldsymbol{X} = \boldsymbol{fusion}(x_1, x_2, x_3). \tag{8}$$

D. Combination of SRFD and DRFD

To assess the efficacy and versatility of our proposed RFD, we conducted experiments on a range of backbone architectures, including transformers, MLP, a mixture of CNNs and transformer, CNNs, and more. We aimed to verify the robustness and generalizability of the RFD across different networks and to validate its potential for practical applications.

We replace the shallow feature DSL (stage 1) of backbone networks with the proposed SRFD module and deep feature DSLs (stages 2–4) with the proposed DRFD module. Specific replacement details for backbone networks are shown in Table I.

The following several DSLs of backbone networks are replaced with our proposed RFD module: GFNet-H [38], swin transformer and its V2 [5], [30], CSWin [31], AS-MLP [36], Mixformer [34], and ConvNeXt [7] (the transformer network has been very popular recently, hence more experiments here). GFNet-H [38] is a backbone network that learns long-term spatial dependencies in the frequency domain with log-linear complexity, which uses a 4 × 4, stride 4 convolution for shallow feature DSL and uses a 2 × 2 convolution with stride 2 for deep feature DSLs. Transformer backbone networks swin transformer and its V2 [5], [30] use a 4 × 4, stride 4 convolution for shallow feature DSL and adopt cut-slice downsampling for deep feature DSLs. Transformer backbone network CSWin [31] uses a 7 × 7, stride 4 convolution for shallow feature DSL and uses a 3 × 3 convolution with stride 2 for deep feature DSLs. MLP backbone network AS-MLP [36] uses a 4 × 4, stride 4 convolution for shallow feature DSL and uses cut-slice downsampling for deep feature DSLs. A mixture of CNNs and transformer backbone network Mixformer [34] uses twice 2 × 2, stride 2 convolution for shallow feature downsampling and adds 2×(3 × 3) convolution with stride 1 between the two convolutional DSLs and uses 2 × 2 convolution with stride 2 for deep feature DSLs. CNNs backbone network ConvNeXt [7] uses a 4 × 4, stride 4 convolution for shallow feature DSL and uses 2 × 2 convolution with stride 2 for deep feature DSLs.

IV. Experiments

In this section, we evaluate the effectiveness of our RFD module on several RS visual tasks. We first compare the RS image classification top-1 accuracy of models with and without our RFD module on the NWPU-RESISC45 [6] dataset. We then ablate the important design elements of our RFD module to compare the degree of importance of the different module designs. We visualize the feature maps corresponding to key structures of our RFD module, such as the feature reinforcement layer and the DSL, to analyze how our RFD module differs from conventional downsampling methods. We also present results of object detection and semantic segmentation on DOTA [16] and iSAID [21] datasets, respectively.

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TABLE I

SRFD AND DRFD MODULE REPLACEMENT DETAILS FOR DSLS OF DIFFERENT BACKBONE NETWORKS. THE NOTATIONS /2 AND /4 INDICATE THE DOWNSAMPLING MULTIPLIER. CONV AND CUT-SLICE DENOTE CONVOLUTIONAL DOWNSAMPLING AND CUT-SLICE DOWNSAMPLING, RESPECTIVELY. THE NETWORKS ARE CLASSIFIED INTO DSL AND FLL

Stage	Output Size	Layer Name	GFNet-H [38]	GFNet-H + RFD	Swins [5], [30]	Swins + RFD	CSWin [31]	CSWin + RFD	AS-MLP [36]	AS-MLP + RFD	Mixformer [34]	Mixformer + RFD	ConvNeXt [7]	ConvNeXt + RFD
stage1	$\frac{H}{4}\times\frac{W}{4}$	DSL	4×4 conv, /4	SRFD	4×4 conv, /4	SRFD	7×7 conv, /4	SRFD	4×4 conv, /4	SRFD	2×2 conv, /2 2∗(3×3 conv)	SRFD	4×4 conv, /4	SRFD
		FLL	GFNet-H	GFNet-H	Swin	Swin	CSWin	CSWin	AS-MLP	AS-MLP	2×2 conv, /2 Mixformer	Mixformer	ConvNeXt	ConvNeXt
stage2	$\frac{H}{8}\times\frac{W}{8}$	DSL	2×2 conv, /2	DRFD	cut-slice, /2	DRFD	3×3 conv, /2	DRFD	cut-slice, /2	DRFD	2×2 conv, /2	DRFD	2×2 conv, /2	DRFD
		FLL	GFNet-H	GFNet-H	Swin	Swin	CSWin	CSWin	AS-MLP	AS-MLP	Mixformer	Mixformer	ConvNeXt	ConvNeXt
stage3	$\frac{H}{16}\times\frac{W}{16}$	DSL	2×2 conv, /2	DRFD	cut-slice, /2	DRFD	3×3 conv, /2	DRFD	cut-slice, /2	DRFD	2×2 conv, /2	DRFD	2×2 conv, /2	DRFD
		FLL	GFNet-H	GFNet-H	Swin	Swin	CSWin	CSWin	AS-MLP	AS-MLP	Mixformer	Mixformer	ConvNeXt	ConvNeXt
stage4	$\frac{H}{32}\times\frac{W}{32}$	DSL	2×2 conv, /2	DRFD	cut-slice, /2	DRFD	3×3 conv, /2	DRFD	cut-slice, /2	DRFD	2×2 conv, /2	DRFD	2×2 conv, /2	DRFD
		FLL	GFNet-H	GFNet-H	Swin	Swin	CSWin	CSWin	AS-MLP	AS-MLP	Mixformer	Mixformer	ConvNeXt	ConvNeXt

TABLE II

TRAINING SETTINGS ON NWPU-RESISC45. T/S/B DENOTE MODELS OF DIFFERENT SIZES

training config	models T/S/B	models + RFD T/S/B
images size	224	224
seed	1234	1234
optimizer	AdamW [43]	AdamW [43]
learning rate	1e-4	1e-4
weight decay	5e-2	5e-2
batch size	16	16
training epochs	300	300
learning rate schedule	cosine decay [44]	cosine decay [44]
warmup	Ture	Ture
warmup epochs	2	2
warmup factor	1e-3	1e-3
RandomResizedCrop	Ture	Ture
RandomHorizontalFlip	Ture	Ture
RandAugment [45]	Ture	Ture

A. Image Classification on NWPU-RESISC45

1) Experimental Settings: We use the NWPU-RESISC45 [6] dataset for RS image scene classification (RESISC). The dataset consists of 31 500 RGB images of size 256 × 256 pixels, collected from Google Earth Imagery in 2017. The images are divided into 45 scene classes, each containing 700 images. The spatial resolution varies from 30.0 to 0.2 m per pixel within the scene classes. We conduct image classification experiments on this dataset, which is also used for ablation studies. To ensure fair comparisons with baseline models and models with our proposed RFD module, we adopt the same training and augmentation strategy. We split the dataset into a training set (80%) and a validation set (20%). We train all models for 300 epochs using the AdamW [43] optimizer with a weight decay of 0.05 and a batch size of 16. We use a linear warmup for the early two epochs and set the initial learning rate to 1e-4 which gradually drops to 1e-3. The data augmentation methods include RandomHorizontalFlip and RandAugment [45]. The specific settings are shown in Table II. All models are trained on an Ubuntu18.04 platform with a GTX3080 GPU. We report top-1 accuracy, parameters, and FLOPs for each model on the NWPU-RESISC45 dataset.

2) Main Results: We replace the DSLs of different backbone networks with our proposed RFD module. The backbone networks include: swin transformer and its V2 [5], [30], pure ViT networks that use Windows multihead self-attention (W-MSA). GFNet-H [38], a network that learns long-term spatial dependencies in the frequency domain with log-linear complexity. AS-MLP [36], an MLP network that uses asymmetric convolutional layers. CSWin [31], a transformer network that computes cross-shaped window self-attention mechanism in horizontal and vertical directions in parallel. Mixformer [34], a mixture of CNNs and transformer networks that leverages both local and global features. ConvNeXt [7], a CNN network designed by borrowing from a swin transformer.

Table III shows the experimental results. We observe that adding our RFD module consistently improves the accuracy of the original models. This demonstrates that our RFD module is suitable for various hierarchical backbone networks, including CNNs, transformer, MLP, and so on. The models equipped with our RFD module achieve varying degrees of improvement in top-1 accuracy, while incurring only a slight change—on average less than 10%—in FLOPs, parameters, and throughput. We conducted throughput tests on a Ubuntu 20.04 platform with a GTX3090Ti GPU. In particular, our Swin V2 + RFD model outperforms the existing state-of-the-art on NWPU-RESISC45 without additional data training. All models obtain different degrees of positive boost by using our RFD module. The transformer backbones get a more significant boost in top-1 accuracy. We did not omit AS-MLP and Mixformer from our results despite their limited improvement. We chose representative backbone networks of different types for our experiments to better demonstrate the robustness and generality of our RFD module.

We conducted a comparison of various downsampling techniques and presented the experimental findings in Table IV. Specifically, we utilized Swin V2 tiny as the baseline model and replaced its deep downsampling module. We trained this model on the NWPU-RESISC45 dataset's training set and reported the top-1 accuracy achieved on the validation set. We compared multiple downsampling modules, including Strided Convolution [27], Maxpooling [1], LIP [39], Antialiased CNNs [40], SoftPool [41], and our DRFD. However, due to the limitations of functions of [1], [39], [40] and [41], their channels could not be modified during downsampling. To address this issue, we made slight modifications

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TABLE III

CLASSIFICATION PERFORMANCE COMPARISON OF INITIAL MODELS AND +RFD MODELS ON NWPU-RESISC45 VALIDATION SET. "+RFD" REPRESENTS THAT THE DSLS OF THE MODEL ARE REPLACED BY THE RFD MODULE. HYBRID DENOTES MIXTURE OF CNN AND TRANSFORMER

Type	Model	Blocks	Channels	initial / +RFD			
				Params (M) ↓	FLOPs (G) ↓	Throughput (fps) ↑	Top-1 (%) ↑
FFT	GFNet-H-T [38]	[3, 3, 10, 3]	[64, 128, 256, 512]	14.60 / 15.68 (+7.4%)	2.05 / 2.43 (+18.5%)	2693 / 2128 (-21.0%)	92.27 / 94.76 (+2.49)
	GFNet-H-S [38]	[3, 3, 10, 3]	[96, 192, 384, 768]	31.43 / 33.05 (+5.2%)	4.59 / 5.34 (+16.3%)	2405 / 2466 (+2.5%)	93.40 / 95.11 (+1.71)
	GFNet-H-B [38]	[3, 3, 27, 3]	[96, 192, 384, 768]	53.01 / 55.43 (+4.6%)	8.53 / 9.28 (+8.8%)	2098 / 1886 (-10.1%)	94.17 / **95.46** (+1.29)
MLP	AS-MLP-T [36]	[2, 2, 6, 2]	[96, 192, 384, 768]	27.55 / 29.96 (+8.7%)	4.39 / 5.14 (+17.1%)	1505 / 1571 (+4.4%)	95.37 / **96.05** (+0.68)
	AS-MLP-S [36]	[2, 2, 18, 2]	[96, 192, 384, 768]	48.86 / 51.27 (+4.9%)	8.57 / 9.32 (+8.8%)	1073 / 1019 (-5.0%)	95.27 / 96.00 (+0.73)
	AS-MLP-B [36]	[2, 2, 18, 2]	[128, 256, 512, 1024]	86.77 / 91.05 (+4.9%)	15.2 / 16.44 (+8.2%)	861 / 830 (-3.6%)	95.63 / 95.94 (+0.31)
Transformer	Swin-T [30]	[2, 2, 6, 2]	[96, 192, 384, 768]	27.55 / 29.97 (+8.8%)	4.36 / 5.11 (+17.2%)	2469 / 2313 (-6.3%)	93.52 / 96.10 (+2.58)
	Swin-S [30]	[2, 2, 18, 2]	[96, 192, 384, 768]	48.87 / 51.28 (+4.9%)	8.52 / 9.27 (+8.8%)	1995 / 1762 (-11.7%)	93.37 / 96.16 (+2.79)
	Swin-B [30]	[2, 2, 18, 2]	[128, 256, 512, 1024]	86.79 / 91.06 (+4.9%)	15.14 / 16.37 (+8.1%)	1975 / 1734 (-12.2%)	93.19 / **96.24** (+3.05)
Transformer	CSWin-T [31]	[1,2,21,1]	[64, 128, 256, 512]	21.83 / 22.05 (+1.0%)	4.08 / 4.35 (+6.6%)	1303 / 1127 (-13.5%)	93.05 / 96.11 (+3.06)
	CSWin-S [31]	[2,4,32,2]	[64, 128, 256, 512]	34.15 / 34.37 (+0.6%)	6.40 / 6.67 (+4.2%)	682 / 579 (-15.1%)	93.56 / 96.11 (+2.55)
	CSWin-B [31]	[2,4,32,2]	[96, 192, 384, 768]	76.65 / 77.12 (+0.6%)	14.36 / 14.87 (+3.6%)	481 / 406 (-15.6%)	94.49 / **96.29** (+1.8)
Transformer	Swin V2-T [5]	[2, 2, 6, 2]	[96, 192, 384, 768]	27.61 / 30.03 (+8.7%)	3.33 / 4.08 (+22.5%)	2009 / 1987 (-1.1%)	94.65 / 96.46 (+1.81)
	Swin V2-S [5]	[2, 2, 18, 2]	[96, 192, 384, 768]	48.99 / 51.41 (+4.9%)	6.47 / 7.22 (+11.6%)	1273 / 827 (-35.0%)	95.22 / **96.84** (+1.62)
	Swin V2-B [5]	[2, 2, 18, 2]	[128, 256, 512, 1024]	86.94 / 91.22 (+4.9%)	11.48 / 12.71 (+10.7%)	1104 / 682 (-38.2%)	95.63 / 96.61 (+0.98)
Hybrid	Mixformer-B0 [34]	[1, 2, 2, 6]	[24, 48, 96, 192]	5.10 / 5.25 (+2.9%)	0.39 / 0.44 (+12.8%)	1287 / 1192 (-7.4%)	94.87 / 95.30 (+0.43)
	Mixformer-B2 [34]	[2, 2, 8, 8]	[32, 64, 128, 256]	9.89 / 10.17 (+2.8%)	0.88 / 0.95 (+8.0%)	1018 / 975 (-4.2%)	95.41 / 96.03 (+0.62)
	Mixformer-B4 [34]	[2, 2, 8, 8]	[64, 128, 256, 512]	34.80 / 35.85 (+3.0%)	3.44 / 3.56 (+3.5%)	830 / 722 (-13.0%)	95.76 / **96.37** (+0.61)
CNN	ConvNeXt-T [7]	[3, 3, 9, 3]	[96, 192, 384, 768]	28.85 / 30.27 (+4.9%)	4.47 / 5.22 (+16.8%)	3109 / 2566 (-17.5%)	93.70 / 95.48 (+1.78)
	ConvNeXt-S [7]	[3, 3, 27, 3]	[96, 192, 384, 768]	49.49 / 51.90 (+4.9%)	8.70 / 9.45 (+8.6%)	2734 / 2478 (-9.4%)	93.90 / 95.48 (+1.58)
	ConvNeXt-B [7]	[3, 3, 27, 3]	[128, 256, 512, 1024]	87.61 / 91.89 (+4.9%)	15.38 / 16.61 (+8.0%)	2726 / 2520 (-7.6%)	95.02 / **96.13** (+1.11)

TABLE IV

COMPARE EXPERIMENTAL RESULTS (TOP1 ACCURACY) OF DEEP DOWNSAMPLING MODULE ON THE NWPU-RESISC45 VALIDATION SET. STRIDED-CONV DENOTES STRIDED CONVOLUTION

Methods	Params(M)	FLOPs(G)	Top-1(%)
Swin V2-T [5]	27.61	3.33	94.65
Swin V2-T + Strided-conv [27]	29.55	3.55	95.11
Swin V2-T + maxpooling [1]	26.45	3.33	95.16
Swin V2-T + LIP [39]	33.42	6.45	95.14
Swin V2-T + antialiased [40]	26.45	3.33	94.24
Swin V2-T + SoftPool [41]	26.45	3.33	94.19
Swin V2-T + DRFD	29.97	3.61	**96.29**

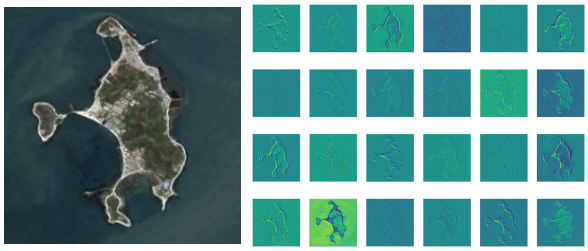


Fig. 5. Visualization of feature maps after feature reinforcement layer in Swin V2-Tiny + RFD.

by first applying a 1 × 1 convolution with stride 1 to adjust the channels to the desired out channels and then downsampling. Our experimental results indicate that our DRFD module achieved significantly higher accuracy than other downsampling modules on the NWPU-RESISC45 dataset.

3) Feature Visualization: Due to RS images often having low resolution and small objects, traditional convolutional downsampling can lead to a loss of detailed features, thereby decreasing the accuracy of the network. As a result, we developed the RFD module to better preserve these features during downsampling. To demonstrate the effectiveness of our approach, we conducted a qualitative analysis that visualized the feature DSL of our network and compared it to the traditional downsampling module.

Fig. 5 shows the feature map of the original image after the feature enhancement layer. From the figure, we can see that the key information of the feature map is preserved and the redundant information is filtered out effectively in the Swin V2-Tiny + RFD network.

Fig. 6 shows the channel data of the 4×, 8×, 16×, and 32× DSLs of the networks. The feature learning blocks of Swin V2 are mainly concentrated in stage 3, the blocks of each stage are [2,2,6,2] or [2,2,18,2], so the feature maps downsampling at 8× to 16×, in stage 3, are crucial. Fig. 6 shows that extracted feature maps of Swin V2 + RFD are more detailed overall than Swin V2 in downsampling from 8× to 16× (only the first 96 channels are listed for each layer feature maps).

Fig. 7 shows an example of a church and medium_ residential, which are buildings with high similarity in the image. The first and third rows are excerpts of downsampled feature maps at different scales of the Swin V2-Tiny network. From the figure, we can see that the network has difficulty distinguishing the features at the 8× scale. The second and fourth rows are excerpts of downsampled features at different scales for the Swin V2-Tiny + RFD network, which replaces the DSLs of Swin V2-Tiny with our proposed RFD module. Compared with the original network, it is obvious that the +RFD model does a better job in texture and detail preservation. We believe this is the most important reason for the higher accuracy of the +RFD model.

For a fair comparison, the test images used were randomly selected from the validation set of NWPU-RESISC45.

B. Object Detection on DOTA

1) Experimental Settings: We conduct object detection experiments on the DOTA 1.0 benchmark [16], which is the

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TABLE V

DETECTION PERFORMANCE COMPARISON OF INITIAL MODELS AND +RFD MODELS ON THE DOTA VALIDATION SET. IT CONTAINS TWO TYPICAL OBJECT DETECTION FRAMEWORKS: ONE-STAGE DECODER S^2ANET AND TWO-STAGE DECODER ORIENTED R-CNN (ORCN). LV: LARGE VEHICLE. SP: SWIMMING POOL. HC: HELICOPTER. BR: BRIDGE. PL: PLANE. SH: SHIP. SBF: SOCCER-BALL FIELD. BC: BASKETBALL COURT. GTF: GROUND TRACK FIELD. SV: SMALL VEHICLE. BD: BASEBALL DIAMOND. TC: TENNIS COURT. RA: ROUNDABOUT. ST: STORAGE TANK. HA: HARBOR

Method	Backbone	mAP (%)	AP per category (%)														
			LV	SP	HC	BR	PL	SH	SBF	BC	GTF	SV	BD	TC	RA	ST	HA
S^2ANet [18]	Swin-T [30]	59.45	75.71	56.68	30.04	32.48	84.89	85.71	51.08	38.90	47.93	60.24	63.54	89.61	48.62	73.44	52.89
	Swin-T + RFD	**66.51**	78.60	61.34	41.64	41.27	88.62	86.65	65.84	55.47	55.40	64.12	68.62	90.55	57.00	79.64	62.87
	ConvNeXt_T [7]	64.31	77.71	59.07	32.64	39.91	85.44	85.57	56.82	48.39	62.48	60.48	68.09	90.41	57.31	76.23	64.02
	ConvNeXt-T + RFD	**68.88**	79.57	63.26	34.82	40.76	89.52	87.13	71.15	66.68	64.34	66.04	71.72	90.76	58.82	83.28	65.29
	AS-MLP-T [36]	67.81	81.92	62.93	42.58	42.40	89.13	88.15	62.50	51.51	63.98	65.62	69.10	90.52	59.60	81.19	66.01
	AS-MLP-T + RFD	**69.04**	81.58	60.54	37.68	45.45	89.93	88.10	68.48	63.10	63.61	68.18	67.37	90.59	61.40	84.39	65.24
ORCN [20]	Swin-T [30]	65.30	81.99	60.66	45.46	35.29	88.62	86.08	56.16	47.88	67.99	59.35	71.12	89.94	50.86	76.99	61.17
	Swin-T + RFD	**70.49**	82.90	64.31	44.60	43.03	90.02	87.62	73.61	62.33	62.57	64.64	74.66	90.76	59.80	83.92	72.62
	ConvNeXt-T [7]	69.14	81.31	60.66	56.43	42.54	89.60	87.16	60.42	56.88	70.77	62.29	71.11	90.35	57.65	78.95	71.02
	ConvNeXt-T + RFD	**71.51**	83.37	64.79	44.28	43.91	90.10	87.44	71.02	66.79	70.93	65.55	72.49	90.77	62.53	85.17	73.47
	AS-MLP-T [36]	69.68	83.58	61.73	41.11	47.16	89.97	87.19	61.29	61.74	71.29	63.32	71.02	90.54	62.08	79.76	73.33
	AS-MLP-T + RFD	**73.01**	83.38	59.68	58.73	47.48	90.00	88.25	73.79	72.01	71.03	65.15	73.49	90.73	62.08	85.54	73.84

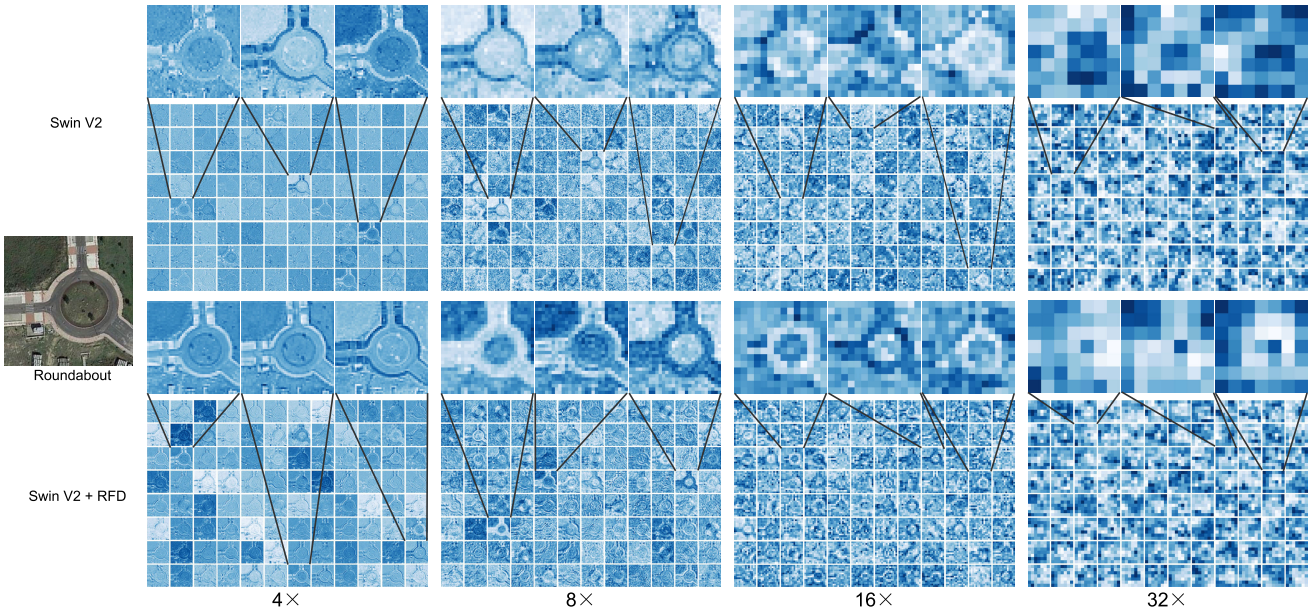


Fig. 6. Visualization of feature maps of 4×, 8×, 16×, and 32× DSLs with different models. Roundabout is the input image. Swin V2 and Swin V2 + RFD denote initial and +RFD models of Swin V2-Tiny, respectively. The test image is randomly selected from the validation set of NWPU-RESISC45, and only the first 96 channels are listed for each layer visualization.

most famous large-scale dataset for oriented bounding box (OBB) detection. It totally contains 2806 images whose size ranges from 800 × 800 to 4000 × 4000, where 188 282 instances belonging to 15 categories are included. The training, validation, and testing set separately have 1411/458/937 tiles. We consider two typical object detection frameworks: one-stage decoder S^2ANet [18] and two-stage decoder Oriented R-CNN (ORCN) [20]. The code is based on Oriented R-CNN [20]. We utilize the single-scale training for the "1×" schedule with 12 epochs. The size of the initial image will be random crop to 1024 × 1024. We take AdamW [43] with a weight decay of 0.05 as the optimizer. The batch size is 2 and the initial learning rate is 1e-3. Other basic settings are the same as Oriented R-CNN. All models are trained on an Ubuntu18.04 platform with a GTX3090Ti GPU.

2) Main Results: Note that the +RFD backbone networks are not pretrained on ImageNet. For a fair comparison, we use the same pretrained weights on the NWPU-RESISC45 training set for all the models. This leads to lower detection accuracy than reported in other papers. Our goal is to verify that the +RFD model can preserve features better than the original model under the same conditions.

Since the object detection codes of swin transformer V2 [5], GFNet-H [38], and Mixformer [34] are not open, the following backbone networks are experimented: MLP backbone network AS-MLP [36], transformer backbone network swin transformer [30], and CNN backbone network ConvNeXt [7]. We report the mAP and all categories of AP of different backbones. Experimental results on the DOTA validation set are shown in Table V. From Table V, it can be seen that regardless of the one-stage or two-stage decoder, backbone networks of the +RFD module make a significant improvement in some categories of AP with a small reduction in some categories of AP, which leads to an overall improvement in the final results.

C. Semantic Segmentation on iSAID

1) Experimental Settings: We evaluate the performance of models via semantic segmentation on the iSAID [21] dataset. iSAID is a large-scale dataset that is mainly for

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TABLE VI

SEGMENTATION PERFORMANCE COMPARISON OF INITIAL MODELS AND +RFD MODELS ON THE iSAID VALIDATION SET. BG: BACKGROUND. SH: SHIP. ST: STORAGE TANK. BD: BASEBALL DIAMOND. TC: TENNIS COURT. BC: BASKETBALL COURT. GTF: GROUND TRACK FIELD. BR: BRIDGE. LV: LARGE VEHICLE. SV: SMALL VEHICLE. HC: HELICOPTER. SP: SWIMMING POOL. RA: ROUNDABOUT. SBF: SOCCER-BALL FIELD. PL: PLANE. HA: HARBOR

Method	Backbone	mIoU (%)	IoU per category (%)															
			BG	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA
UperNet [47]	Swin-T [30]	51.00	98.39	51.91	50.26	60.3	79.57	28.68	37.97	18.10	55.51	31.38	22.53	46.13	57.52	53.87	75.82	48.10
	Swin-T + RFD	**57.65**	98.64	58.40	58.50	72.97	86.27	51.07	44.88	23.04	56.71	37.05	25.95	48.16	62.17	67.10	78.00	53.42
	ConvNeXt-T [7]	48.33	98.38	50.09	45.12	68.85	78.3	27.59	46.49	19.21	52.48	24.74	3.20	41.21	38.87	58.83	71.09	48.87
	ConvNeXt-T + RFD	**52.22**	98.48	51.14	55.44	71.16	83.54	44.06	48.53	30.48	50.07	28.57	3.00	44.85	39.51	64.90	72.87	48.93
	AS-MLP-T [36]	52.65	98.46	53.76	56.99	64.92	83.43	47.85	43.64	27.88	55.23	30.99	3.77	41.85	52.47	57.99	75.66	47.56
	AS-MLP-T + RFD	**55.65**	98.54	56.77	61.26	72.33	83.29	51.78	54.39	32.37	55.05	30.01	9.92	46.59	49.39	67.04	72.05	49.52
SegNeXt [48]	Swin-T [30]	49.45	98.31	49.14	46.96	55.93	73.63	24.39	39.23	19.26	55.95	28.08	22.48	43.37	59.42	53.87	73.97	47.21
	Swin-T + RFD	**57.27**	98.62	55.89	57.83	69.11	83.41	49.74	48.78	27.10	58.09	36.15	24.20	47.06	66.40	67.16	76.28	50.56
	ConvNeXt-T [7]	46.52	98.29	47.27	42.99	66.79	74.90	24.48	46.29	17.69	52.20	22.03	0.49	36.96	44.20	57.31	69.75	42.76
	ConvNeXt-T + RFD	**53.44**	98.49	50.06	53.58	70.49	81.49	47.02	49.62	31.38	54.71	30.82	0.25	46.84	52.94	64.30	73.76	49.37
	AS-MLP-T [36]	54.18	98.52	54.57	54.38	67.76	82.52	51.01	50.48	28.68	57.14	30.71	1.56	41.22	61.80	62.92	74.75	48.83
	AS-MLP-T + RFD	**58.62**	98.62	58.00	62.13	73.78	82.78	55.09	57.64	33.56	57.76	33.42	24.45	45.08	61.02	67.74	76.10	50.72

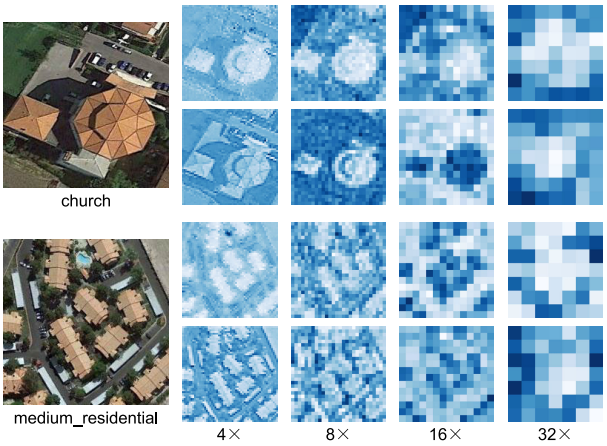


Fig. 7. Visualization of feature maps of 4×, 8×, 16×, and 32× DSLs with Swin V2-Tiny init and +RFD models. Church and medium_residential are input images. The first and third rows are excerpts of downsampled feature maps at different scales of the Swin V2-Tiny network. The second and fourth rows are excerpts of downsampled features at different scales for the Swin V2-Tiny + RFD network.

instance segmentation. Also, it provides the semantic mask including 15 foregrounds and one background category over the aerial objects. It consists of 2806 high-resolution images that range from 800 × 800 to 4000 × 13 000 pixels. The training, validation, and test sets separately have 1411, 458, and 937 images, respectively. In this article, only the validation set is used for evaluation since the testing set is unavailable. We crop the original images to 1024 × 1024 and then resize them to 512 × 512, containing a total of 11 534 training images and 3949 validation images. We follow the default settings of mmsegmentation [48]. We take UperNet [46] and SegNeXt [47] as the main frameworks and take different architectures as backbones. We take AdamW [43] with a weight decay of 0.01 as an optimizer for 80 K iterations. The learning rate is 6e-5 with 1500 iteration warmup at the beginning of training and linear learning rate decay. The augmentations include random flipping, random scaling, and random photometric distortion. The input batch size is set as 4, and single test. All models are trained on an Ubuntu18.04 platform with a GTX3090Ti GPU.

2) Main Results: Same as object detection, for a fair comparison, we use the same pretrained weights on the NWPU-RESISC45 training set for all the models. This leads to lower segmentation accuracy than reported in other papers.

The following backbone networks are experimented: the MLP backbone network AS-MLP [36], transformer backbone network swin transformer [30], and CNN backbone network ConvNeXt [7]. We chose two decoders, UperNet [46] and SegNeXt [47], for a total of 12 experiments in six combinations. The results are reported in Table VI. Experimental results show that the addition of the proposed RFD significantly improves the mIoU of original models. In particular, it can be seen from Table VI that almost every category of IoU is positively boosted by using the RFD module in the backbone networks, demonstrating the robustness of the RFD module.

TABLE VII

ABLATION EXPERIMENTAL RESULTS (TOP1 ACCURACY) OF MODELS AFTER ADDING THE SRFD OR DRFD MODULE ON THE NWPU-RESISC45 DATASET WITH 20% TRAINING RATE

SRFD	DRFD	Params(M)	FLOPs(G)	Top-1(%)
		27.61	3.33	85.89
✓		27.67	3.81	86.65
	✓	29.97	3.61	89.89
✓	✓	30.03	4.08	**90.45**

D. Ablation Study

All ablation experimental results are conducted on NWPU-RESISC45 that split 20% data for training and 80% data for validation. The Swin V2-Tiny model is selected for all the ablation experiments. We report experimental results with singlecrop top-1 accuracy, parameters, and FLOPs.

1) RFD Ablation: We conduct ablation experiments on the SRFD module and the DRFD module. Experimental results are shown in Table VII. The last row is our RFD module. Experimental results show that the model with the RFD module added achieves the best top-1 accuracy.

2) SRFD Ablation: We test the feature reinforcement layer on the SRFD module, which tests the convolutional kernels of sizes 3 × 3, 5 × 5, 7 × 7, and 9 × 9, it makes a consistent improvement in model performance. The experimental results show that the kernel size 7 × 7 is the best. Furthermore, we test adding GELU activation [42] layer on the SRFD module and the results show that the accuracy of Swin V2-Tiny is decreased. Therefore, we do not add the GELU layer to the SRFD module.

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TABLE VIII

ABLATION EXPERIMENTAL RESULTS (TOP1 ACCURACY) OF THE DRFD MODULE (WITHOUT SRFD) ON THE NWPU-RESISC45 DATASET WITH 20% TRAINING RATE. D_{conv}, D_{max}, AND D_{cut} DENOTE CONVOLUTION, MAXPOOLING, AND CUT-SLICE DOWNSAMPLING, RESPECTIVELY

D_{conv}	D_{max}	D_{cut}	Params(M)	FLOPs(G)	Top-1(%)
✓			26.09	3.17	86.73
	✓		26.08	3.17	88.96
		✓	27.61	3.33	85.89
	✓	✓	29.18	3.52	89.25
✓		✓	29.19	3.52	87.60
✓	✓		27.64	3.35	89.42
	DRFD		29.97	3.61	**89.89**

3) DRFD Ablation: We conduct the DRFD module ablation experiments (without SRFD), and the experimental results are shown in Table VIII. Experimental results show that the model with the DRFD module added achieved the best top-1 accuracy.

V. CONCLUSION

In this work, we propose a new plug-and-play and effective downsampling module called RFD that consists of SRFD and DRFD. We argue that convolutional downsampling can fuse local information, but it may also cause blurring or loss of detailed information about objects. This is especially detrimental to RS images, which are characterized by smaller and more objects. Detailed information is crucial for RS visual tasks. Therefore, our RFD module aims to preserve the features better than conventional convolutional downsampling. We evaluate our RFD module on various types of backbone networks (e.g., CNN, transformer, MLP, and a mixture of CNN and transformer) and conduct extensive comparison experiments on several public RS image datasets for classification, object detection, and semantic segmentation tasks. The consistent performance improvements demonstrate the effectiveness and generalizability of our RFD module.

In the future, we will further explore the RFD module in more diverse visual tasks. We hope that this work can inspire more research on improving the feature extraction in DSLs, instead of just focusing on FLLs of models.

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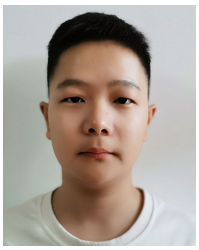
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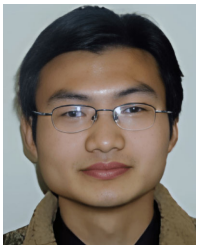
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